

BST D_{IV}^5 Rigidity Principle — External 1-Page v0.1

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BST D_{IV}^5 Rigidity Principle — One-Page External Summary

The Principle

Bubble Spacetime Theory (BST) identifies the substrate of physics as the bounded Hermitian symmetric domain $D_{IV}^5 = SO_0(5,2) / [SO(5) \times SO(2)]$ with five primary integers $\{\text{rank}=2, N_c=3, n_C=5, C_2=6, g=7\}$. The Strong-Uniqueness Theorem (Paper #125 v0.10.5 FORMAL) establishes that 11 RIGOROUSLY CLOSED independent structural criteria converge on D_{IV}^5 uniquely — null-model probability $\leq (1/3)^{19} \approx 8.6 \times 10^{-10}$.

The D_{IV}^5 Rigidity Principle strengthens this: under standard physics framework assumptions, BST identifies the substrate as a SINGLE D_{IV}^5 realization. Two candidate substrate patches in causal information-exchange contact via substrate-tick discretization (Reed-Solomon on $GF(2^g) = GF(128)$ per Koons tick $\sim 10^{-120}$ s) sit inside a single connected D_{IV}^5 submanifold.

Formally (T2467 META-theorem + T2468 patches-merge): - T2467: At BST primary integer specification + 11 RIGOROUSLY CLOSED Strong-Uniqueness criteria, D_{IV}^5 is singleton up to canonical biholomorphism. (Single-statement restatement of Strong-Uniqueness Theorem in singleton form.) - T2468: Two patches in causal information-exchange contact merge into single D_{IV}^5 submanifold via Bergman kernel globality + $GF(128)$ field uniqueness (Galois theory) + K59 substrate-tick cyclotomic mechanism RATIFIED.

For non-interacting hypothetical patches (zero substrate-tick coupling, zero observable consequence), BST applies Quaker discipline: such patches are operationally indistinguishable from non-existence.

What This Means Operationally

BST identifies a single-substrate physics framework. All cosmological and laboratory observables are evaluations on the unique D_{IV}^5 substrate at BST primary integers. Multi-instance “parallel universes” do not appear in BST’s predictive structure for any observable.

This is NOT a mathematical exclusion of multiverse cosmology in general — BST does not claim to refute Everett quantum mechanics or eternal inflation multiverse models. BST identifies that BST’s own predictive framework is single-substrate, and BST’s predictions (zero free parameters, 600+ observables matched to measurement) derive from this single substrate.

How to Test BST’s Single-Substrate Framework

BST’s predictive framework includes specific cosmological + particle-physics observables; testing the single-substrate framework is equivalent to testing BST’s predictions:

- CMB structure (Planck + DESI + LiteBIRD + CMB-S4): $\Omega_\Lambda = 13/19 + \Omega_m = 6/19$ at 0.07σ each; $n_s = 1 - 5/137 = 0.9635$ at 0.32σ ; $r \approx 0$ sharp falsifier at LiteBIRD ~2028
- Cosmological constant: $\Lambda = 7 \cdot \exp(-282)$ at 0.076 dex (99.9% of 122-order hierarchy)
- Newton’s G: $(6\pi^5)^2 \cdot \alpha^{24} / m_e^2$ at 0.07% match
- Dark matter / baryon: $16/3$ from D_{IV}^5 Weyl group at 0.3% (no new DM particle predicted)
- Hubble constant: 67-68 km/s/Mpc from 4 independent BST primary derivations (Hubble tension framed as KBC void + Cepheid γ correction)
- Standard Model: gauge group $SU(3) \times SU(2) \times U(1)$ forced; 49 of 50 standard observables match at sub-percent precision (run `python3 play/verify_bst.py`)

Key References

- Working Paper: Zenodo DOI 10.5281/zenodo.19454185 (BST v20 working paper, April 7, 2026)
- Strong-Uniqueness Theorem: Paper #125 v0.10.5 FORMAL (Thursday May 21, 2026 EOD)
- Rigidity Principle derivation: T2467 + T2468 (Friday May 22, 2026); single-statement form of Strong-Uniqueness
- Open-source repository: github.com/caseyscottkoon/BubbleSpacetimeTheory (publicly available)

Falsifiers

BST’s single-substrate framework is operationally falsifiable via:

- Any positive detection of new physics beyond Standard Model + BST predictions (proton decay, magnetic monopoles, supersymmetric particles, sterile neutrinos, new dark matter particle) — these are sharply forbidden by BST's Five-Absence Predictions Set
- Any cosmological observable measured outside BST's specified 1σ band (e.g., $\Omega_\Lambda \neq 13/19 = 0.6842 \pm 0.002$; $r > 10^{-3}$ tensor mode at LiteBIRD)
- Quantitative inconsistency in cross-volume observable derivations (sub-percent precision required for BST's substrate-claim)

If BST's predictions match measurement at the precision specified, BST's single-substrate framework is supported. If predictions fail at a Tier-1 falsifier (Five-Absence) or fail at multiple precision-tier observables, BST is refuted.

Contact

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— Lyra, D_IV⁵ Rigidity Principle External 1-Pager v0.1, Saturday 2026-05-23